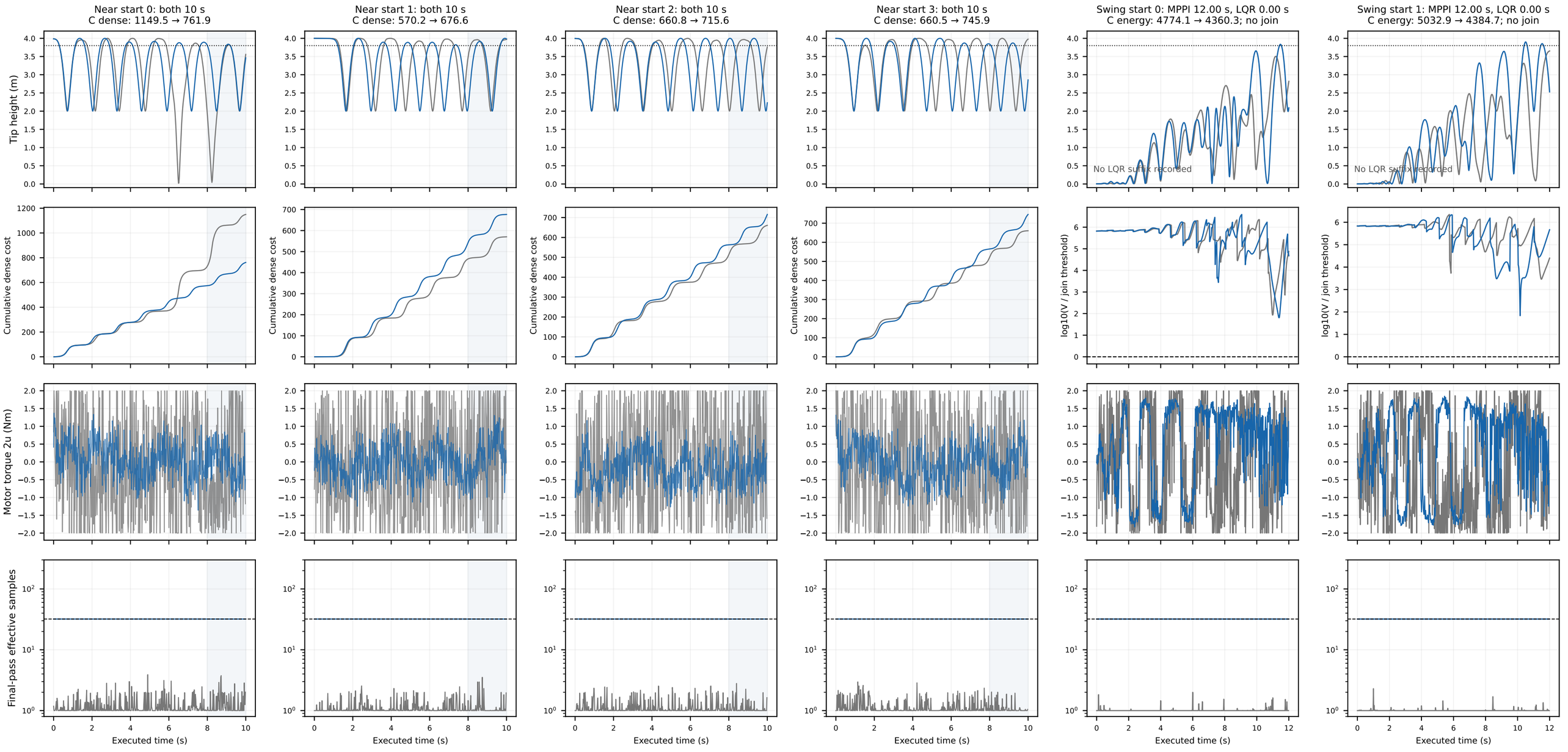

Controller evidence: temperature changes weight spread; basin entry remains a separate physical outcome
Near start 0: both 10 s
C dense: 1149.5 → 761.9
Near start 1: both 10 s
C dense: 570.2 → 676.6
Near start 2: both 10 s
C dense: 660.8 → 715.6
Near start 3: both 10 s
C dense: 660.5 → 745.9
Swing start 0: MPPI 12.00 s, LQR 0.00 s
C energy: 4774.1 → 4360.3; no join
Swing start 1: MPPI 12.00 s, LQR 0.00 s
C energy: 5032.9 → 4384.7; no join
No LQR suffix recorded
No LQR suffix recorded
Tip height (m)
Cumulative dense cost
log10(V / join threshold)
Motor torque 2u (Nm)
Final-pass effective samples
Executed time (s)
Gray: historical λ=3 (near) or λ=.15 (swing); blue: adaptive ESS32 MPPI; green: LQR only if recorded. Near cost/duration are identical definitions.
Near dense and swing energy objectives have different definitions/scales; totals are not ranked across them. ESS is undefined during LQR. No new physics.